Wentao Yuan

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Education

University of Washington, Seattle, WA 2019 – 2024
Ph.D. in Computer Science
Advisor: Dieter Fox

Carnegie Mellon University, Pittsburgh, PA 2017 – 2019
M.S. in Robotics
Advisor: Martial Hebert

Pomona College, Claremont, CA 2013 – 2017
B.A. in Computer Science and Mathematics (Summa Cum Laude)

Publications

Wentao Yuan, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithya Murali, Arsalan Mousavian, Dieter Fox. RoboPoint: A Vision-Language Model for Spatial Affordance Prediction for Robotics. In *Conference on Robot Learning (CoRL)*, 2024.

Jiafei Duan, **Wentao Yuan**, Wilbert Pumacay, Yi Ru Wang, Kiana Ehsani, Dieter Fox, and Ranjay Krishna. Manipulate-Anything: Automating Real-World Robots using Vision-Language Models. In *Conference on Robot Learning (CoRL)*, 2024.

Adam Fishman, Aaron Walsman, Mohak Bhardwaj, **Wentao Yuan**, Balakumar Sundaralingam, Byron Boots, and Dieter Fox. Avoid Everything: Model-Free Collision Avoidance with Expert-Guided Fine-Tuning. In *Conference on Robot Learning (CoRL)*, 2024.

Wentao Yuan, Adithya Murali, Arsalan Mousavian, Dieter Fox. M2T2: Multi-Task Masked Transformer for Object-centric Pick and Place. In *Conference on Robot Learning (CoRL)*, 2023.

Xiangyun Meng, Nathan Hatch, Alexander Lambert, Anqi Li, Nolan Wagener, Matthew Schmittle, JoonHo Lee, **Wentao Yuan**, Zoey Chen, Samuel Deng, Greg Okopal, Dieter Fox, Byron Boots, Amirreza Shaban. TerrainNet: Visual Modeling of Complex Terrain for High-speed, Off-road Navigation. In *Robotics: Science and Systems (RSS)*, 2023.

Wentao Yuan, Chris Paxton, Karthik Desingh, Dieter Fox. SORNet: Spatial Object-Centric Representations for Sequential Manipulation. In *Conference on Robot Learning (CoRL)*, 2021 (**oral**). **Best Systems Paper Finalist**.

Wentao Yuan, Zhaoyang Lv, Tanner Schmidt, Steven Lovegrove. STaR: Self-supervised Tracking and Reconstruction of Rigid Objects in Motion with Neural Rendering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Ben Eckart, **Wentao Yuan**, Chao Liu, Jan Kautz. Self-Supervised Learning on 3D Point Clouds by Learning Latent Generative Models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Wentao Yuan, Ben Eckart, Kihwan Kim, Varun Jampani, Dieter Fox, Jan Kautz. DeepGMR: Learning Latent Gaussian Mixture Models for Registration. In *European Conference on Computer Vision (ECCV)*, 2020 (**spotlight**).

Wentao Yuan, David Held, Christoph Mertz, Martial Hebert. Iterative Transformer Network for 3D Point Cloud. In *CVPR Workshop on 3D Scene Understanding for Vision, Graphics, and Robotics*, 2019.

Wentao Yuan, Tejas Khot, David Held, Christoph Mertz, Martial Hebert. PCN: Point Completion Network. In *International Conference on 3D Vision (3DV)*, Verona, Italy, 2018 (**oral**). **Best Paper Honorable Mention**.

Chahyon Ku, Carl Winge, Ryan Diaz, **Wentao Yuan**, Karthik Desingh. Evaluating Robustness of Visual Representations for Object Assembly Task Requiring Spatio-Geometrical Reasoning. In *International Conference on Robotics and Automation (ICRA)*, 2024.

Jeffery Rutledge*, **Wentao Yuan***, Jane Wu, Zoe Wood, Timmy Gambin, Christopher Clark. Intelligent Shipwreck Search Using Autonomous Underwater Vehicles. In *International Conference on Robotics and Automation (ICRA)*, 2018.

Preprints

Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, **Wentao Yuan**, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. AHA: A Vision-Language-Model for Detecting and Reasoning Over Failures in Robotic Manipulation. *arXiv preprint arXiv:2410.00371*, 2024.

Experience

Google DeepMind, Mountain View, CA — 10/2024 – present
Research Scientist
Manager: Jie Tan

NVIDIA, Seattle, WA — 02/2021 – 09/2024
Research Intern
Mentor: Adithya Murali, Arsalan Mousavian
Project: Scalable Robot Manipulation in Cluttered Environments

Meta Reality Labs Research, Redmond, WA — 07/2020 – 11/2020
Research Intern
Mentor: Tanner Schmidt, Zhaoyang Lv
Project: Neural Rendering of Dynamic Scenes

NVIDIA, Santa Clara, CA — 06/2019 – 12/2019
Research Intern
Mentor: Ben Eckart, Kihwan Kim
Project: Learnable 3D Gaussian Mixture Models

Google, New York, NY — 05/2016 – 08/2016
Software Engineering Intern

Google, Kirkland, WA — 05/2015 – 08/2015
Engineering Practicum Intern

Reviewer

Conferences

- CoRL since 2022
- RSS since 2024
- ICRA since 2023
- IROS since 2021
- CVPR since 2023
- ICCV since 2023
- ECCV since 2024
- NeurIPS since 2023
- ICML since 2023
- ICLR since 2024

Journals

- IEEE RA-L
- IEEE T-RO
- IEEE T-PAMI
- IJCV
- VISI